

AP Statistics FRQ Practice Exam 02

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AP Statistics FRQ Practice Exam 2

APSTAT-FRQ2-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: None

FRQ Type: Formulate Questions and Collect Data

Prompt:

A city museum has 8,400 adult members in a database ordered by the date each person joined. The museum wants to learn whether members support adding evening hours. An employee chooses a random integer from 1 through 70 and then selects that member and every 70th member after that in the database. The resulting 120 selected members receive an email survey. Forty-seven members respond as shown.

Response	Number
Yes	32
No	15
Total	47

The email asks, “Wouldn’t you agree that convenient new evening hours should be offered?”

(a) Identify the population and the intended sample for the survey.

(b) Identify the sampling method used to select the 120 members. Explain the role of the random starting integer.

(c) Calculate the sample proportion among respondents who answered “Yes.” Identify the population parameter this statistic is intended to estimate, and state a statistical question the museum could investigate.

(d) Explain how the wording of the email question could create response bias. Rewrite the question in a more neutral form.

(e) Explain how nonresponse could bias the result even though the 120 members were selected using a probability method. Describe one practical step the museum could take to reduce or assess nonresponse bias.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population as all 8,400 adult members in the museum's database.
- Point 2: Identifies the intended sample as the 120 members selected and contacted, while recognizing that the observed response group contains only the 47 respondents.

(b) 1 point

- Point 1: Identifies a systematic random sample and explains that the random start gives database positions a chance to enter the every-70th-member selection pattern; an equivalent explanation of avoiding a fixed, employee-chosen start is acceptable.

(c) 3 points

- Point 1: Calculates $\hat{p} = 32/47 \approx 0.681$, or 68.1%, using the number of respondents rather than all 120 contacted members as the denominator.
- Point 2: Identifies the parameter p as the true proportion of all 8,400 adult museum members who support adding evening hours.
- Point 3: States an appropriate statistical question, such as, "What proportion of the museum's adult members support adding evening hours?" or "Do more than half of the museum's adult members support adding evening hours?"

(d) 2 points

- Point 1: Explains that "Wouldn't you agree" and "convenient" favor a Yes response and may lead respondents rather than measure their opinions neutrally.
- Point 2: Gives a neutral revision, such as, "Do you support or oppose adding evening hours at the museum?" with balanced response choices.

(e) 2 points

- Point 1: Explains that the 73 nonrespondents may differ systematically from the 47 respondents in their support for evening hours, so 68.1% may overestimate or underestimate the population proportion.
- Point 2: Proposes a relevant step, such as repeated follow-up contacts, offering mail or telephone response options, or randomly sampling nonrespondents for intensive follow-up and comparing their answers with those of initial respondents.

Model Guidance:

The target population is the museum's entire adult membership, and the intended sample consists of the 120 database entries selected by a random-start systematic procedure. The 47 respondents are the people from whom measurements were actually obtained. Choosing the first position at random prevents the employee from deliberately choosing where the selection pattern begins. A hidden pattern that repeats every 70 database positions could still be a concern, although ordering only by join date does not by itself establish such a pattern.

Among respondents, $\hat{p} = 32/47 \approx 0.681$ estimates the proportion p of all members who support evening hours. That estimate is vulnerable to two separate errors. The wording is leading because it labels the proposal "convenient" and invites agreement. A neutral version should name the proposal without signaling a preferred answer. In addition, the low response rate means the respondents may not represent the selected members. Follow-ups or a study of randomly chosen nonrespondents would help reduce or measure that possible nonresponse bias.

Expected Points:

APSTAT-FRQ2-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: Unit 3, Inference for Categorical Data: Proportions

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

A regional electric-bike sharing service operates 12,000 battery packs. Based on stable long-term records, 18% of the packs require sensor recalibration at their annual inspection. An auditor takes a simple random sample of 150 packs. Let $\hat{p}$ be the proportion in the sample that require recalibration.

- (a) Check the conditions needed to model the sampling distribution of $\hat{p}$ as approximately normal.
- (b) Calculate the mean and standard deviation of the sampling distribution of $\hat{p}$. Interpret the standard deviation in context.
- (c) Sketch the approximate sampling distribution of $\hat{p}$ on paper. Label its center and the value 0.12, and shade the region representing $P(\hat{p} \leq 0.12)$.
- (d) Calculate $P(\hat{p} \leq 0.12)$. Interpret the probability in context.
- (e) Suppose the auditor instead takes an SRS of 600 packs from the same fleet. Calculate the new standard deviation and approximate $P(\hat{p} \leq 0.12)$. Explain why this probability differs from the result for $n = 150$.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Cites the SRS and verifies the 10% condition because $150 \leq 0.10(12,000)$, supporting approximate independence.
- Point 2: Verifies the large-counts condition: $np = 150(0.18) = 27$ and $n(1 - p) = 150(0.82) = 123$, both at least 10.

(b) 2 points

- Point 1: Calculates $\mu_{\hat{p}} = p = 0.18$ and $\sigma_{\hat{p}} = \sqrt{[0.18(0.82)/150]} \approx 0.03137$.
- Point 2: Interprets 0.03137 as the typical sample-to-sample distance between $\hat{p}$ and the population proportion 0.18 for repeated SRSs of 150 packs.

(c) 2 points

- Point 1: Draws an approximately normal, bell-shaped curve for $\hat{p}$ centered at 0.18, with a horizontal axis labeled as sample proportion requiring recalibration.
- Point 2: Marks 0.12 to the left of 0.18 and shades the area at or below 0.12.

(d) 2 points

- Point 1: Standardizes and calculates $z = (0.12 - 0.18)/0.03137 \approx -1.91$ and $P(\hat{p} \leq 0.12) \approx 0.0279$.
- Point 2: Interprets the result: if the population proportion is 0.18, about 2.79% of SRSs of 150 packs would have 12% or fewer requiring recalibration.

(e) 2 points

- Point 1: Calculates $\sigma\hat{p} = \sqrt{[0.18(0.82)/600]} \approx 0.01568$, $z \approx -3.83$, and $P(\hat{p} \leq 0.12) \approx 0.00007$.
- Point 2: Explains that a larger sample size reduces the standard deviation of $\hat{p}$, so 0.12 is farther below the mean in standard-deviation units and is much less likely when $p = 0.18$.

Model Guidance:

Because the sample is random, is far below 10% of the fleet, and has expected counts of 27 recalibration cases and 123 noncases, a normal model is appropriate. The sampling distribution is centered at the population proportion 0.18, with spread $\sqrt{[p(1 - p)/n]}$. For $n = 150$, that spread is about 0.03137. A sample result of 0.12 is therefore about 1.91 standard deviations below the center, giving a lower-tail probability near 0.0279.

The sketch should show a normal curve centered at 0.18 and a small shaded left tail ending at 0.12. When n increases to 600, the standard deviation is halved to about 0.01568 because the sample size is multiplied by 4. The same cutoff is then about 3.83 standard deviations below the center, so its probability falls to approximately 0.00007. The population proportion has not changed; only the sampling variability has.

Expected Points:

10

APSTAT-FRQ2-003

Section: FRQ

Type: frq

Primary Unit: Unit 4, Inference for Quantitative Data: Means

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

An agricultural cooperative is evaluating software that automatically assists with planning drone flight paths. From the 240 fields served by the cooperative, 12 fields are selected using a simple random sample. For each selected field, the same trained technician prepares one flight plan manually and one with software assistance. The order of the two methods is randomized for each field.

For each field, define d = assisted planning time – manual planning time, measured in minutes. The paired differences have the following summary statistics.

$$n = 12$$

$$\bar{x}_d = -2.40 \text{ minutes}$$

$$sd = 1.70 \text{ minutes}$$

A dotplot of the 12 differences is roughly symmetric, with no apparent outliers. For 11 degrees of freedom, the critical value for a 95% confidence interval is $t^* = 2.201$.

- (a) Identify the population parameter to be estimated and name the appropriate confidence-interval procedure.
- (b) Check the conditions for the confidence-interval procedure.
- (c) Calculate the standard error and margin of error, and construct a 95% confidence interval for the population mean difference.
- (d) Interpret the 95% confidence interval in context.
- (e) A cooperative manager claims that software assistance reduces mean planning time by more than 1 minute for fields served by the cooperative. Use the interval to evaluate the claim.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies μ_d as the true mean of assisted-minus-manual planning-time differences for all 240 fields served by the cooperative.
- Point 2: Selects a paired t confidence interval for a population mean difference, rather than an independent two-sample procedure.

(b) 3 points

- Point 1: Cites the simple random sample of fields and notes that the paired differences from distinct fields can reasonably be treated as independent.
- Point 2: Verifies the 10% condition because $12 \leq 0.10(240)$.
- Point 3: Uses the described roughly symmetric dotplot with no apparent outliers to justify a t procedure despite the small sample size.

(c) 3 points

- Point 1: Calculates $SE = sd/\sqrt{n} = 1.70/\sqrt{12} \approx 0.4907$ minute.
- Point 2: Calculates the margin of error as $t^* \times SE = 2.201(0.4907) \approx 1.08$ minutes.
- Point 3: Constructs -2.40 ± 1.08 , giving the interval $(-3.48, -1.32)$ minutes, allowing for minor rounding differences.

(d) 1 point

- Point 1: States that the researchers are 95% confident that the true mean assisted-minus-manual planning-time difference for all fields served by the cooperative is between -3.48 and -1.32 minutes, equivalent to a mean reduction between 1.32 and 3.48 minutes with assistance.

(e) 1 point

- Point 1: Supports the manager's claim because the entire interval for assisted minus manual is below -1 minute; all plausible values from the interval correspond to a mean reduction greater than 1 minute.

Model Guidance:

The same field is measured under both methods, so the observational units for inference are the 12 within-field differences. The parameter is μ_d , where negative values favor assistance. The random selection of fields, the sample being less than 10% of the population, and the roughly symmetric,

outlier-free distribution of differences support a paired t interval.

The estimated standard error is $1.70/\sqrt{12} \approx 0.4907$ minute. With $t^* = 2.201$, the margin of error is about 1.08 minutes, and the interval is $(-3.48, -1.32)$. Thus, plausible values indicate that software assistance reduces mean planning time by 1.32 to 3.48 minutes. Because even the endpoint closest to zero represents a reduction greater than 1 minute, the interval supports the manager's claim for the cooperative's fields.

Expected Points:

10

APSTAT-FRQ2-004

Section: FRQ

Type: frq

Primary Unit: Unit 5, Regression Analysis

Secondary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A park district takes a simple random sample of 60 mature silver maple trees from the district's inventory. For each tree, x is trunk diameter in centimeters and y is canopy area in square meters. The observed trunk diameters range from 15 to 55 centimeters.

Statistic	x : trunk diameter	y : canopy area
Mean	34 cm	48 m ²
Standard deviation	10 cm	15 m ²
Correlation	$r = 0.80$	

For trees at a fixed trunk diameter within the observed range, the regression residuals are adequately modeled by a normal distribution with mean 0 and standard deviation 9.0 m².

(a) Use the summary statistics to calculate the slope and intercept of the least-squares regression line for predicting canopy area from trunk diameter. State the equation.

(b) For a tree with a trunk diameter of 40 cm, calculate the standardized value z_x , the predicted standardized canopy value $z_{\hat{y}} = rz_x$, and the predicted canopy area. Explain how the two standardized values illustrate regression toward the mean.

(c) According to the model, calculate the probability that a mature silver maple with a trunk diameter of 40 cm has a canopy area of at most 65 m². Interpret the probability.

(d) Without using trunk diameter, predicting 48 m² for every sampled tree gives a typical error measured by $s_y = 15$ m². With trunk diameter, the residual standard deviation is 9 m². Evaluate the claim that using the regression model reduces typical prediction error by at least one-third.

(e) The district arborist claims that at least 85% of its mature silver maples with trunk diameters near 40 cm have canopy areas no greater than 65 m². Use the statistical evidence and study design to evaluate the claim, and state an important condition on your conclusion.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Calculates $b_1 = r(s_y/s_x) = 0.80(15/10) = 1.20$ m² per centimeter.
- Point 2: Calculates $b_0 = \bar{y} - b_1\bar{x} = 48 - 1.20(34) = 7.20$ m² and states $\hat{y} = 7.20 + 1.20x$.

(b) 2 points

- Point 1: Calculates $z_x = (40 - 34)/10 = 0.60$, $z\hat{y} = 0.80(0.60) = 0.48$, and $\hat{y} = 48 + 0.48(15) = 55.2$ m².
- Point 2: Explains that because $|r| < 1$, the predicted response is 0.48 standard deviation above its mean, closer to 0 than the predictor's 0.60 standard deviation; this movement toward the response mean is regression toward the mean.

(c) 2 points

- Point 1: Calculates the residual cutoff $65 - 55.2 = 9.8$ m², standardizes it as $z = 9.8/9.0 \approx 1.09$, and obtains $P(y \leq 65 \mid x = 40) = P(Z \leq 1.09) \approx 0.862$.
- Point 2: Interprets the result: under the regression and normal-residual model, about 86.2% of mature silver maples with trunk diameter 40 cm are expected to have canopy areas at most 65 m².

(d) 2 points

- Point 1: Calculates the relative reduction in typical prediction error as $(15 - 9)/15 = 0.40$, or 40%.
- Point 2: Concludes that the observed reduction is greater than one-third and therefore supports the claim for these data, while recognizing that s_y describes mean-only prediction error and the residual standard deviation describes regression prediction error.

(e) 2 points

- Point 1: States that the model estimate 0.862 exceeds 0.85, so the evidence supports, but does not prove with certainty, the arborist's "at least 85%" claim.
- Point 2: Limits the conclusion to mature silver maples in the park district with diameters near 40 cm and states that it depends on the linear regression and constant normal-residual model being reasonable there; the SRS supports generalization to the district's inventory.

Model Guidance:

The slope can be recovered from standardized units: $b_1 = r s_y/s_x = 1.20$, and the intercept makes the line pass through $(\bar{x}, \bar{y}) = (34, 48)$, producing $\hat{y} = 7.20 + 1.20x$. At $x = 40$, the predictor is 0.60 standard deviation above its mean. Multiplication by r gives a predicted response only 0.48 standard deviation above its mean, or 55.2 m². This shrinkage in standardized distance is regression toward the mean.

For a 40 cm tree to have canopy area at most 65 m², its residual must be at most $65 - 55.2 = 9.8$ m². Under the stated normal residual model, this is $z \approx 1.09$ and a probability of about 0.862. The reduction from a typical error of 15 m² to 9 m² is 40%, which exceeds one-third. Together, the model-based probability and random sampling support the arborist's claim for comparable district trees, provided the linear model and the stated residual distribution are appropriate near 40 cm. The value 0.862 is a model-based long-run proportion, not a guarantee about every group of such trees.

Expected Points:

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ2

- APSTAT-FRQ2-001: Unit 1 primary; no secondary unit; systematic random sample from 8,400 adult city-museum members using a random start and every 70th database entry, with 120 contacted and 47 respondents; Yes/No response table with 32 Yes and 15 No; identify population and intended sample, calculate respondent $\hat{p} = 32/47$, define the population proportion, formulate a support question, diagnose leading wording and nonresponse bias, and propose neutral wording plus follow-up; targets confusion between probability selection and an unbiased final response set.
- APSTAT-FRQ2-002: Unit 2 primary; Unit 3 secondary; sampling distribution of $\hat{p}$ for annual sensor recalibration among 12,000 electric-bike battery packs with $p = 0.18$; SRS sizes 150 and 600; normal-curve sketch centered at 0.18 with left tail at 0.12; check random/10%/large-count conditions, calculate and interpret $\sigma\hat{p}$, find lower-tail probabilities 0.0279 and about 0.00007, and explain the sample-size effect; targets confusing the population proportion with sampling variability and assuming a larger sample changes the center.
- APSTAT-FRQ2-003: Unit 4 primary; Unit 1 secondary; paired t confidence interval for assisted-minus-manual drone flight-planning time on an SRS of 12 from 240 cooperative fields, with randomized method order, $\bar{x}_d = -2.40$ minutes, $sd = 1.70$, a roughly symmetric outlier-free dotplot description, and $t^* = 2.201$; identify μ_d and pairing, check SRS/independence/10%/shape conditions, calculate SE, margin of error, and 95% interval $(-3.48, -1.32)$, interpret the interval, and assess a mean reduction exceeding 1 minute; targets using an independent two-sample method and reversing the sign of the defined difference.
- APSTAT-FRQ2-004: Unit 5 primary; Unit 2 secondary; SRS of 60 mature silver maples with trunk diameter x and canopy area y , summary-statistics table $\bar{x} = 34$, $s_x = 10$, $\bar{y} = 48$, $s_y = 15$, $r = 0.80$, diameter range 15–55, and normal residual SD 9.0; derive $\hat{y} = 7.20 + 1.20x$, use standardized regression at $x = 40$ to explain regression toward the mean, calculate $P(y \leq 65 \mid x = 40) \approx 0.862$, compare 15 m² mean-only error with 9 m² regression error for a 40% reduction, and evaluate an at-least-85% district claim subject to sampling scope and model assumptions; targets confusing correlation with slope, missing regression toward the mean, and treating a model probability as certainty.